

ECON 609: Value Function Iteration

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Lecture # 3

Introduction

- ▶ Introduce numerical methods to solve dynamic programming (DP) models.
- ▶ DP models with sequential decision making:
 - ▶ Arrow, Harris, and Marschak (1951): optimal inventory model.
 - ▶ Lucas and Prescott (1971): optimal investment model.
 - ▶ Brock and Mirman (1972): optimal growth model under uncertainty
 - ▶ Lucas (1978): asset pricing models.
 - ▶ Kydland and Prescott (1982): business cycle model.

The Basic Framework

- ▶ Almost any DP can be formulated as Markov Chain decision process (MDP).
- ▶ An agent, given state $s_t \in S$ takes an optimal action $a_t \in A(s)$ that determines current utility $u(s_t, a_t)$ and affects the distribution of next period's state s_{t+1} via a Markov chain $p(s_{t+1}|s_t, a_t)$.
- ▶ The problem is to choose $\alpha = \{\alpha_1, \dots, \alpha_T\}$, where $a_t = \alpha_t(s_t)$, that solves

$$V(s) = \max_{\alpha} \mathbb{E} \left\{ \sum_{t=0}^T \beta^t u(s_t, a_t) \mid s_0 = s \right\}$$

The DP Problem

- ▶ DP simplifies the MPD problem, allowing us to find $\alpha = \{\alpha_1, \dots, \alpha_T\}$ using a recursive procedure.
- ▶ We are going to focus on infinite horizon problems, where V is the unique solution for the Bellman equation $V = \Gamma(V)$.
- ▶ Γ is called the Bellman operator, that is defined as:

$$\Gamma(V)(s) = \max_a [u(s, a) + \beta \int V(s') p(ds'|s, a)]$$

- ▶ $\alpha(s)$ is equal to the solution to the Bellman equation for $\forall s$.

The Bellman Operator and the Bellman Equation

- ▶ We will revisit the mathematical foundations for the Bellman equation.
- ▶ It has a very nice property: Γ is a contraction mapping.
- ▶ This will allow us to use some numerical procedures to find the solution to the Bellman equation recursively.

Computation: Speed vs. Accuracy

- ▶ The approximating error ϵ introduces a trade-off: better accuracy vs. shorter time to find the solution.
- ▶ The time needed to find the solution also depends on the dimension of the problem d .
- ▶ We want the fastest method given a pair (ϵ, d) .
- ▶ Normally, these algorithms are nested into a bigger optimization algorithm.
- ▶ Hence, we will have to solve the Bellman equation for various values of the structural parameters defining β , u , and p .

Approximation to Continuous DPs

- ▶ There are two ways to approximate continuous DPs whose state and control variables take a continuum of values.
 - ▶ Discrete
 - ▶ Smooth
- ▶ Discrete solves an equivalent discrete problem that approximates the original continuous DPs.
- ▶ Smooth treats the value function V and the decision rule α are smooth functions of s and a finite set of parameters θ .

The Bellman Equation in the Finite Horizon Problem

- ▶ If $T < \infty$ (the problem has a finite horizon), DP is equivalent to backward induction. In the terminal period α_T is:

$$\alpha_T(s_T) = \arg \max_{a_T \in A(s_T)} u(s_T, a_T)$$

- ▶ And $V_T(s_T) = u(s_T, \alpha_T(s_T))$.
- ▶ For periods $t = 1, \dots, T - 1$, we can find V_t and α_t by

$$\alpha_t(s_t) = \arg \max_{a_t \in A(s_t)} \left[u(s_t, a_t) + \beta \int V_{t+1}(s_{t+1}) p(ds_{t+1} | s_t, a_t) \right]$$

$$V_t(s_t) = u(s_t, \alpha_t(s_t)) + \beta \int V_{t+1}(s_{t+1}) p(ds_{t+1} | s_t, \alpha_t(s_t))$$

- ▶ Markovian property of p implies that $a_t = \alpha_t(s_t)$ instead of depending on the whole history.

The Bellman Equation in the Infinite Horizon Problem I

- ▶ If $T = \infty$, we do not have a finite state.
- ▶ Markovian property of p implies that the problem has a stationary Markovian structure.
- ▶ The optimal Markovian rule is characterized by

$$\alpha(s) = \arg \max_{a \in A(s)} \left[u(s, a) + \beta \int V(s') p(ds' | s, a) \right]$$
$$V(s) = u(s, \alpha(s)) + \beta \int V(s') p(ds' | s, \alpha(s))$$

- ▶ This equation is known as the Bellman equation.
- ▶ The function V is the fixed point to this functional equation.

The Bellman Equation in the Infinite Horizon Problem II

- ▶ To determine existence and uniqueness, we need to impose

- (1) S and A are compact metric spaces.
- (2) $u(s, a)$ is jointly continuous and bounded.
- (3) $s \rightarrow A(s)$ is a continuous correspondence.

- ▶ Let $B(S)$ the Banach space of bounded, measurable real-valued functions on S .

- ▶ Let $\|f\| = \sup_{s \in S} |f(s)|$ for $f \in B(S)$ be the sup norm.

- ▶ The Bellman operator is

$$\Gamma(W)(s) = \max_{a \in A(s)} \left[u(s, a) + \beta \int W(s') p(ds' | s, a) \right]$$

- ▶ The Bellman equation is a fixed point to $V = \Gamma(V)$.

The Bellman Equation in the Infinite Horizon Problem III

- ▶ Blackwell (1965) and Denardo (1967) show that the Bellman operator is a contraction mapping: for W, V in $B(S)$,

$$||\Gamma(V) - \Gamma(W)|| \leq \beta ||V - W||$$

- ▶ Contraction mapping theorem: if Γ is a contraction mapping on a Banach space B , then Γ has a unique fixed point.
- ▶ Blackwell's theorem: the stationary Markovian α defined by

$$\alpha(s) = \arg \max_{a \in A(s)} \left[u(s, a) + \beta \int V(s') p(ds' | s, a) \right]$$

$$V(s) = u(s, \alpha(s)) + \beta \int V(s') p(ds' | s, \alpha(s))$$

solves the associated MDP problem.

Phelps' (1972) Example

- ▶ The agent has to decide between consume and save.
- ▶ The state variable w is the wealth of the agent and the decision variable c is how much to consume.
- ▶ The agent cannot borrow, so the choice set $A(w) = \{c | 0 \leq c \leq w\}$.
- ▶ The savings are invested in a single risky asset with i.i.d. return R with distribution F .
- ▶ The Bellman equation?

Numerical Implementation

- ▶ We reviewed some theoretical background on dynamic programming.
- ▶ Now we will discuss its numerical implementation.
- ▶ Perhaps the most important solution algorithm to learn:
 - (1) Wide applicability
 - (2) Many known results
 - (3) Template for other algorithms
- ▶ Importance of keeping the curse of dimensionality under control.

Numerical Implementation

- ▶ We have tight convergence properties and bounds on errors.
- ▶ It will always (perhaps quite slowly) work.
- ▶ How do we implement the operator?
 - (1) We come back to our two distinctions: finite vs. infinite time & discrete vs. continuous state space.
 - (2) Then we need to talk about initialization and discretization.

Value Function Iteration in Finite Time

- ▶ We begin with the Bellman operator:

$$\Gamma(V^t)(s) = \max_{a \in A(s)} \left[u(s, a) + \beta \int V^{t'}(s') p(ds'|s, a) \right]$$

- ▶ Specify V^T and apply Bellman operator:

$$V^{T-1}(s) = \max_{a \in A(s)} \left[u(s, a) + \beta \int V^T(s') p(ds'|s, a) \right]$$

- ▶ Iterate until the first period:

$$V^1(s) = \max_{a \in A(s)} \left[u(s, a) + \beta \int V^s(s') p(ds'|s, a) \right]$$

Value Function Iteration in Infinite Time

- ▶ We begin with the Bellman operator:

$$\Gamma(V)(s) = \max_{a \in A(s)} \left[u(s, a) + \beta \int V(s') p(ds'|s, a) \right]$$

- ▶ Specify V^0 and apply Bellman operator:

$$V^1(s) = \max_{a \in A(s)} \left[u(s, a) + \beta \int V^0(s') p(ds'|s, a) \right]$$

- ▶ Iterate until convergence:

$$V^T(s) = \max_{a \in A(s)} \left[u(s, a) + \beta \int V^{T-1}(s') p(ds'|s, a) \right]$$

Normalization

- ▶ Before initializing the algorithm, it is usually a good idea to normalize the problem:

$$V(s) = \max_{a \in A(s)} \left[(1 - \beta) u(s, a) + \beta \int V(s') p(ds' | s, a) \right]$$

- ▶ This normalization is known to have better stability properties.

Initial Value in Finite Time Problems

- ▶ Usually, economics of the problem provides natural choices.
- ▶ Example: final value of an optimal expenditure problem is zero.
- ▶ However, sometimes there are subtle issues.
- ▶ Example: what is the value of dying? And of bequest? OLG models.

Initial Value in Infinite Time Problems

- ▶ Theorems tell us we will converge from any initial guess.
- ▶ This does not mean that we should not be smart picking our initial guess.
- ▶ Several good ideas:
 - ▶ Steady state of the problem (if it exists).
 - ▶ Perturbation approximation.
 - ▶ Collapsing one or more dimensions of the problem. Which one?

Discretization

- ▶ In case we have a continuous state space, we need to discretize it into a grid.

- ▶ Exact problem:

$$V(s) = \max_{a \in A(s)} \left[(1 - \beta)u(s, a) + \beta \int V(s')p(ds'|s, a) \right]$$

- ▶ Approximated problem:

$$\hat{V}(s) = \max_{a \in A(s)} \left[(1 - \beta)u(s, a) + \beta \sum_{k=1}^N \hat{V}(s'_k)p_N(s'_k|s, a) \right]$$

Grid Generation

- ▶ Huge literature on numerical analysis on how to efficiently generate grids.
- ▶ Two main issues:
 - (1) How to select points s_k
 - (2) How to approximate p by p_N
- ▶ Answer to second issue follows from answer to first problem.
- ▶ We can combine strategies to generate grids.

Uniform Grid

- ▶ Decide how many points in the grid.
- ▶ Distribute them uniformly in the state space.
- ▶ What if the state space is not bounded?
- ▶ Advantages and disadvantages.

Discretizing Stochastic Process: Tauchen's Method

- ▶ Importance case: discretizing exogenous stochastic process.
- ▶ Consider a general AR(1) process:

$$z' = (1 - \rho)\mu_z + \rho z + \epsilon, \quad \epsilon \sim_{i.i.d} \mathcal{N}(0, \sigma_\epsilon^2)$$

- ▶ Recall that $\mathbb{E}[z] = \mu_z$ and $\text{Var}[z] = \sigma_z^2 = \frac{\sigma_\epsilon^2}{(1-\rho^2)}$.
- ▶ First step is to choose m (e.g., $m = 3$) and N , and define:

$$z_N = \mu_z + m\sigma_z, \quad z_1 = \mu_z - m\sigma_z$$

- ▶ $z_2, \dots, z_{N-1}$ are equispaced over the interval $[z_1, z_N]$ with $z_k < z_{k+1}$ for any $k \in \{1, \dots, N-1\}$.

Transition Probability

- ▶ Once we have the grid points, how do we compute the transition probabilities?

Transition Probability

- ▶ Let $d = z_{k+1} - z_k$. Then

$$\begin{aligned}\pi_{i,j} &= \Pr\{z' = z_j | z = z_i\} \\&= \Pr\{z_j - d/2 < z' \leq z_j + d/2 | z = z_i\} \\&= \Pr\{z_j - d/2 < (1 - \rho)\mu_z + \rho z_i + \epsilon \leq z_j + d/2\} \\&= \Pr\left\{\frac{z_j - d/2 - (1 - \rho)\mu_z - \rho z_i}{\sigma_\epsilon} < \frac{\epsilon}{\sigma_\epsilon} \leq \frac{z_j + d/2 - (1 - \rho)\mu_z - \rho z_i}{\sigma_\epsilon}\right\} \\&= \Phi\left(\frac{z_j + d/2 - (1 - \rho)\mu_z - \rho z_i}{\sigma_\epsilon}\right) - \Phi\left(\frac{z_j - d/2 - (1 - \rho)\mu_z - \rho z_i}{\sigma_\epsilon}\right)\end{aligned}$$

- ▶ Adjust for tails:

$$\pi_{i,j} = \begin{cases} 1 - \Phi\left(\frac{z_j - d/2 - (1 - \rho)\mu_z - \rho z_i}{\sigma_\epsilon}\right) & \text{if } j = N \\ \Phi\left(\frac{z_j + d/2 - (1 - \rho)\mu_z - \rho z_i}{\sigma_\epsilon}\right) - \Phi\left(\frac{z_j - d/2 - (1 - \rho)\mu_z - \rho z_i}{\sigma_\epsilon}\right) & \text{otherwise} \\ \Phi\left(\frac{z_j + d/2 - (1 - \rho)\mu_z - \rho z_i}{\sigma_\epsilon}\right) & \text{if } j = 1 \end{cases}$$

Interpolation

- ▶ Discretization also generates the need for interpolation.
- ▶ Simpler approach: linear interpolation.
- ▶ Problem: curse of dimensionality, linear interpolation may not preserve shape (e.g. differentiability, concavity).
- ▶ Shape-preserving splines: Schumaker scheme.

Applying the Algorithm

- ▶ After deciding initialization and discretization, we still need to implement each step:

$$V^T(s) = \max_{a \in A(s)} \left[u(s, a) + \beta \int V^{T-1}(s') p(ds'|s, a) \right]$$

- ▶ Two numerical operations:
 - (1) Maximization: brute force, Newton method, simulated annealing...
 - (2) Integral: exact integration, quadrature, Monte Carlo...

Convergence Assessment

- ▶ How do we assess convergence?
- ▶ Stop at iteration k if $\|V^{k+1} - V^k\| < tol$, where tol is a chosen tolerance level.
- ▶ We can also conduct non-local accuracy test proposed by Judd (1992) and Judd and Guu (1997).
- ▶ Example: Euler equation from a stochastic neoclassical growth model

$$\frac{1}{c(k_t, z_t)} = \mathbb{E}_t \left(\frac{\beta \alpha e^{z_{t+1}} k(k_t, z_t)^{\alpha-1}}{c(k(k_t, z_t), z_{t+1})} \right)$$

We can define the following error:

$$EE(k_t, z_t) \equiv 1 - c(k_t, z_t) \mathbb{E}_t \left(\frac{\beta \alpha e^{z_{t+1}} k(k_t, z_t)^{\alpha-1}}{c(k(k_t, z_t), z_{t+1})} \right)$$

Appendix

(only when we have enough time)

Endogenous Grid Method

- ▶ Proposed by **Carroll (2005)**.
- ▶ It is actually easier to understand with a concrete example: a basic stochastic neoclassical growth model.
- ▶ The problem has a Bellman equation representation:

$$V(k_t, z_t) = \max_{k_{t+1}} \left\{ \frac{(e^{z_t} k_t^\alpha + (1 - \delta)k_t - k_{t+1})^{1-\sigma}}{1 - \sigma} + \beta \mathbb{E}_t V(k_{t+1}, z_{t+1}) \right\}$$
$$\text{s.t. } z_{t+1} = \rho z_t + \epsilon_{t+1}$$

Changing State Variables

- ▶ We will use a state variable called "market resources" or "cash-on-hand", instead of k_t :

$$Y_t = c_t + k_{t+1} = y_t + (1 - \delta)k_t = e^{z_t} k_t^\alpha + (1 - \delta)k_t$$

- ▶ We use a capital Y_t to denote the total market resources and a lower y_t for the production function.
- ▶ Change of variables serves as a key in solving our problem.
- ▶ As a result, we write the problem with the Bellman equation:

$$V(Y_t, z_t) = \max_{k_{t+1}} \left\{ \frac{(Y_t - k_{t+1})^{1-\sigma}}{1 - \sigma} + \beta \mathbb{E}_t V(Y_{t+1}, z_{t+1}) \right\}$$

s.t. $z_{t+1} = \rho z_t + \epsilon_{t+1}$

Optimality Condition

- ▶ Since Y_{t+1} is only a function of k_{t+1} and z_{t+1} , we can write

$$\tilde{V}(k_{t+1}, z_t) \equiv \beta \mathbb{E}_t V(Y_{t+1}, z_{t+1})$$

to get

$$V(Y_t, z_t) = \max_{k_{t+1}} \left\{ \frac{(Y_t - k_{t+1})^{1-\sigma}}{1-\sigma} + \tilde{V}(k_{t+1}, z_t) \right\}$$

- ▶ The first order condition is

$$(c_t^*)^{-\sigma} = \tilde{V}_{k_{t+1}}(k_{t+1}^*, z_t)$$

where $c_t^* = Y_t - k_{t+1}^*$.

Backing Up Consumption

- ▶ If we know $\tilde{V}(k_{t+1}, z_t)$, then $c_t^* = \left(\tilde{V}_{k_{t+1}}(k_{t+1}, z_t) \right)^{-\frac{1}{\sigma}}$ for each point in a grid for k_{t+1} and z_t .

- ▶ Given c_t^* and k_{t+1} , we can find $Y_t^* = c_t^* + k_{t+1}$ and obtain

$$V(Y_t^*, z_t) = \left\{ \frac{(c_t^*)^{1-\sigma}}{1-\sigma} + \tilde{V}(k_{t+1}, z_t) \right\}$$

where we drop the max operator, since we have already computed the optimal level of consumption.

- ▶ Since $Y_t^* = e^{z_t}(k_t^*)^\alpha + (1-\delta)k_t^*$, an alternative interpretation of the algorithm is that, during the iterations, the grid on k_{t+1} is fixed, but the values of k_t change endogenously.

Comparison with Standard Approach

- ▶ In the standard VFI, the optimality condition is

$$(c_t^*)^{-\sigma} = \beta \mathbb{E}_t V_{k_{t+1}}(k_{t+1}^*, z_{t+1})$$

- ▶ Since $c_t = e^{z_t} k_t^\alpha + (1 - \delta)k_t - k_{t+1}$, we have to solve

$$(e^{z_t} k_t^\alpha + (1 - \delta)k_t - k_{t+1}^*)^{-\sigma} = \beta \mathbb{E}_t V_{k_{t+1}}(k_{t+1}^*, z_{t+1})$$

a nonlinear equation on k_{t+1}^* for each point in a grid for k_t .

- ▶ The key difference is that the endogenous grid method defines a fixed grid over the values of k_{t+1} instead of over k_t .
- ▶ We already know what values the policy function for next period's capital take and, thus, we can skip the root-finding.